Leftover Bytes

(End User Insights & Market Analysis)

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User Interview Findings

Students

(12 Interviews)

Demographics

Primarily ages 18-25, diverse dietary needs

Behaviours

1. Typically unaware of leftover food availability until it's too late
2. Seeks easy access to food alerts via mobile notifications
3. Prefer diet-specific food options (e.g., vegetarian, gluten-free)
4. Motivated by convenience and instant availability
5. Interested in gamified features like rewards for sharing photos or claiming food

Pain Points

1. Misses out on available food due to lack of timely updates
2. Only engages when food is nearby and easily accessible

Event Organizers

(2 Interviews)

Demographics

Event coordinators, staff, or faculty managing events

Behaviours

1. Frequently has excess food after campus events
2. Seeks a quick, user-friendly tool to manage food sharing
3. Open to community-driven updates (sharing status, food availability)
4. Looks for ways to engage students more effectively and increase turnout

Pain Points

1. Struggles with connecting students to leftover food
2. Lacks an efficient method to handle food waste reduction post-event

What We Heard: Student Voices

Key Features (Leftover Bytes)	Student Voices
Food Access Pain Point	*"I've seen so much food wasted after events, but I never know when or where it happens."*
Notification Preference	*"Real-time alerts on my phone would be way better than Slack or email."*
Transparency Matters	*"Sometimes I'd go and the food was already gone. If people updated the status, that'd help."*
Dietary Needs	*"I'd love to get alerts based on my diet – I'm vegetarian and hate seeing irrelevant alerts."*
Location-Based Usefulness	*"I'll only go if it's within like 5 minutes. Timing and proximity really matter."*
Gamification Feedback	*"Free food + points for photos? I'm definitely doing that."*

Google Form Responses

QR Code for Form

What would motivate you the most to participate in a leftover food sharing system?

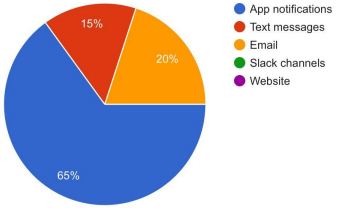
7 responses

- Money saved in groceries and time saved in cooking, it also helps that the leftover food is generally of good quality and mostly healthy
- Some sort of reward system (like earning points)
- Something to get in return
- Some goodies or participation certificates
- It's good not to waste food
- Controlling the food wastage
- The shouldnt be a leftover but should be free :)

How would you want to be notified if free food if available on campus?

Copy chart

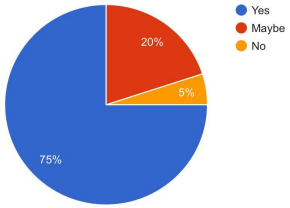
59 responses



would seeing a picture of leftover food help you decide whether to get it?

Copy chart

59 responses



Have you ever picked up leftover food from an event

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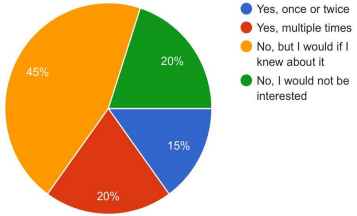
59 responses



would you be willing to post a quick photo or update for other students after picking up leftover food?

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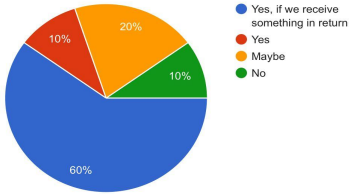
59 responses



JTBD Framework

Primary User – BU Student

Functional Jobs

- Get real-time alerts about available leftover food near their location.
- Filter notifications based on dietary preferences (e.g., veg, halal).
- Know how recently food was posted and how far it is.
- Optionally update others about food status to earn rewards.

Emotional Jobs

- Feel included in resource-sharing opportunities.
- Reduce guilt or stress associated with food insecurity.
- Feel motivated and rewarded for contributing to a sustainable community.

Social Jobs

- Gain recognition through leaderboards and certificates.
- Be seen as a contributor to sustainability and zero-waste initiatives.

Secondary User – Event Organizer

Social Jobs

- Be seen as responsible and sustainability-oriented within the BU community.

Emotional Jobs

- Feel helpful and impactful without overextending time or resources.
- Trust the platform to handle logistics after their part is done.

Functional Jobs

- Easily list leftover food with minimal effort post-event.
- Ensure real-time communication with potential student recipients.
- Track participation without staying back or handling manual distribution.

SWOT Analysis - Leftover Bytes

Strengths

- Mobile-first experience enhances user accessibility.
- Real-time alerts & photo updates keep users informed instantly.
- Dietary filters make the app inclusive and user-friendly.
- Gamified features (leaderboards, points, certificates, goodies) boost user engagement.
- Tackles food waste in a sustainable and tech-forward way.

Weakness

- Still in prototype phase – no MVP launched yet.
- Requires development and iterative testing.
- User adoption unknown – no data yet on usability or interest.

Opportunities

- Sustainability trend: More institutions and students are eco-conscious.
- Partner with BU: Expand reach and gain institutional support.
- Room for features: Meal history, ratings, volunteer integration, etc.
- Educational potential: Could be used to promote campus sustainability awareness.

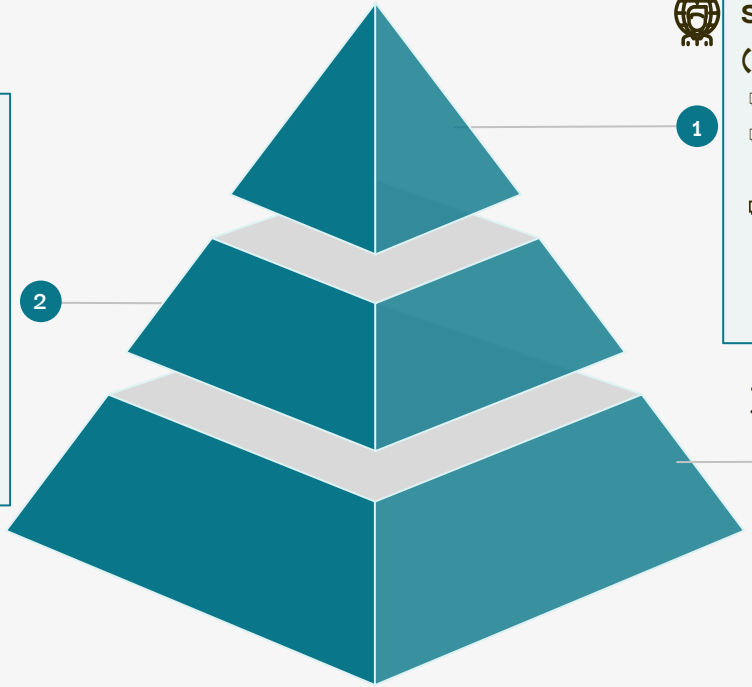
Threats

- Requires BU buy-in: Potential delays due to institutional approval.
- Culture shift: Students may be slow to adopt new behavior around leftover food.
- Competing platforms: If similar apps exist or BU has other initiatives.
- Privacy or health concerns around sharing leftover food may arise.

Feature Grid Analysis

Features	OLIO	Too Good To Go	Slack Channel (BU)	Feed Share (NEU)	Leftover Bytes (Our App)
Real-time Notifications	❌	✅	❌	✅	✅
Dietary Filters	❌	❌	❌	❌	✅
Campus Specific Availability	❌	❌	✅	✅	✅
Photo Uploads from Users	✅	✅	❌	❌	✅
Leaderboard & Certifications	❌	❌	❌	❌	✅
Gamification (points system & goodies) & Rewards	❌	❌	❌	❌	✅
Free to use	✅	❌	✅	✅	✅

Market Sizing (TAM, SAM, SOM)



Serviceable Obtainable Market (SOM)

- App downloads and **active users in Year 1**
- Based on app awareness, gamification, and social sharing
- **Est. 15% of SAM = ~1,300 users**
  ~1,000 passive users (notification → pickup)
  ~200–300 highly engaged users (post, earn points)

Serviceable Available Market (SAM)

- App for students attending on-campus events with leftover food available
- Based on interest in free food, diet-filtered notifications, and reducing waste
- **Est. 70% of TAM actively participate = ~8,820 students**

Total Addressable Market (TAM)

- **Total students at BU: ~36,000** (undergrad + grad)
- Target segment: Food-insecure or budget/sustainability-conscious students
- Est. 35% of students fall into this group
- **TAM = ~12,600 students**

ROI & Potential for Return

Institutional Impact
(Return for BU)

- Supports **BU's Zero Waste** & Sustainability goals by reducing food waste.
- Makes events **more efficient** and appreciated—leftovers don't go to waste.
- Offsets operational costs—**no extra staff/logistics** needed for redistribution.
- Boosts **student engagement** in green initiatives with minimal overhead.

Community Value
(Return for Students)

- **Free food** access for food-insecure & budget-conscious students.
- **Rewards system** (leaderboard, certificates, goodies) encourages participation.
- Promotes **student-led sustainability** and peer accountability.
- Enhances **campus belonging** and sharing-oriented culture.

Sustainability
(Environmental ROI)

- **Saves 120–200 portions/week** in pilot (4 CDS events/day).
- Reduces over 100 lbs of food waste/week, conserving water & energy.
- **Minimizes trash overflow**, improving cleanup and waste management.

Thank You!